HW1

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Q1: A.3

Let $I = [0, 1]$, $\Omega = \mathbb{R}$ and for $\alpha \in \mathbb{R}$, $A_\alpha = (\alpha - 1, \alpha + 1)$, the open interval $\{x : \alpha - 1 < x < \alpha + 1\}$.

(a)

Show that $\bigcup_{\alpha \in I} A_\alpha = (-1, 2)$ and $\bigcap_{\alpha \in I} A_\alpha = (0, 1)$.

Answer

Sketch: To prove equality we argue both directions for both the Union and Intersection proofs, i.e. $\subset$ and $\supset$.

Note: we use $\subset \equiv \subseteq$ and $\supset \equiv \supseteq$ for the purposes of set relations, as noted by Dr. Ghosh in lecture.

Union

Proof.

($\subset$) For any $\alpha \in [0, 1]$, $\alpha - 1 \in [-1, 0]$ and $\alpha + 1 \in [1, 2]$, so $A_\alpha = (\alpha - 1, \alpha + 1) \subset (-1, 2)$. Since this holds for every $\alpha \in I$,

$$\bigcup_{\alpha \in I} A_\alpha \subset (-1, 2).$$

($\supset$) Let $x \in (-1, 2)$ be given, and set $\alpha = \min(\max(x, 0), 1) \in [0, 1] = I$.

- If $x < 0$: $\alpha = 0$, and since $-1 < x < 0 < 1$, $x \in A_0 = (-1, 1)$.
- If $0 \leq x \leq 1$: $\alpha = x$, and trivially $x - 1 < x < x + 1$, so $x \in A_x$.
- If $x > 1$: $\alpha = 1$, and since $0 < 1 < x < 2$, $x \in A_1 = (0, 2)$.

In every case $x \in A_\alpha$ for some $\alpha \in I$, so $x \in \bigcup_{\alpha \in I} A_\alpha$. Hence

$$(-1, 2) \subset \bigcup_{\alpha \in I} A_\alpha.$$

Combining both containments, we conclude $\bigcup_{\alpha \in I} A_\alpha = (-1, 2)$.//

Intersection

Proof.

($\subset$) Since $0, 1 \in I$, $\bigcap_{\alpha \in I} A_\alpha \subset A_0 \cap A_1 = (-1, 1) \cap (0, 2) = (0, 1)$.

($\supset$) Let $x \in (0, 1)$ be given, and let $\alpha \in I$ be arbitrary. Since $\alpha \geq 0$, $\alpha - 1 \geq -1$. And since $0 \leq \alpha \leq 1$, it follows that $\alpha - 1 \leq 0 < x$ and $x < 1 \leq \alpha + 1$, for $x \in A_\alpha$. As $\alpha \in I$ was arbitrary, $x \in \bigcap_{\alpha \in I} A_\alpha$. Hence

$$(0,1) \subset \bigcap_{\alpha \in I} A_\alpha.$$

Combining both containments, we conclude $\bigcap_{\alpha \in I} A_\alpha = (0,1)$.//

(b)

Suppose $J = \{x : x \in I, x \text{ is rational}\}$. Find $\bigcup_{x \in J} A_x$ and $\bigcap_{x \in J} A_x$.

Answer

Sketch: The key is that 0 and 1 are each rational, so $0, 1 \in J$; this lets us use the results from part (a) to force the reverse containments.

Union

Proof. Since $J \subset I$,

$$\bigcup_{x \in J} A_x \subset \bigcup_{\alpha \in I} A_\alpha = (-1,2).$$

Conversely, since $0, 1 \in J$ (both are rational numbers in I),

$$\bigcup_{x \in J} A_x \supset A_0 \cup A_1 = (-1,1) \cup (0,2) = (-1,2),$$

using that the two intervals overlap on $(0,1)$. Combining both containments,

$$\bigcup_{x \in J} A_x = (-1,2).//$$

Intersection

Since $J \subset I$,

$$\bigcap_{x \in J} A_x \supset \bigcap_{\alpha \in I} A_\alpha = (0,1).$$

Conversely, since $0, 1 \in J$,

$$\bigcap_{x \in J} A_x \subset A_0 \cap A_1 = (-1,1) \cap (0,2) = (0,1).$$

Combining both containments,

$$\bigcap_{x \in J} A_x = (0,1).//$$

Q2: A.5

Show that $X \equiv \{0,1\}^{\mathbb{N}}$, the set of all sequences $\{\omega_i\}_{i \in \mathbb{N}}$ where each $\omega_i \in \{0,1\}$, is **uncountable**. Conclude that $\mathcal{P}(\mathbb{N})$ is uncountable.

Answer

Sketch: To prove $X \equiv \{0,1\}^{\mathbb{N}}$ is uncountable, and that consequently $\mathcal{P}(\mathbb{N})$ is uncountable, we first establish (1): X is uncountable and then (2): $\mathcal{P}(\mathbb{N})$ and X are in bijection.

(1): X is uncountable

Suppose, for contradiction, that X is countable. Since X is countable and nonempty (it contains, e.g., the constant sequence $\omega_i = 0$ for all i), $\exists$ a surjection $h : \mathbb{N} \to X$ enumerating every element of X: $h(1), h(2), h(3), \ldots$

Write $h(n) = (h(n)_1, h(n)_2, h(n)_3, \ldots)$ for the n-th sequence in the list, with $h(n)_i \in \{0,1\}$ its i-th coordinate.

Define $\delta = (\delta_i)_{i\in\mathbb{N}}$ by flipping the diagonal entries: $\delta_i = 1 - h(i)_i, \qquad i \in \mathbb{N}$.

Then $\delta \in X$, since each $\delta_i \in \{0,1\}$. But for every $n \in \mathbb{N}$, $\delta_n = 1 - h(n)_n \neq h(n)_n$, so δ and $h(n)$ disagree in coordinate n, i.e. $\delta \neq h(n)$. This holds for **every** n, so $\delta \notin \{h(1), h(2), h(3), \ldots\}$, contradicting the assumption that h is surjective (enumerates all of X).

Hence no such enumeration exists, and X is uncountable.

(2): $\mathcal{P}(\mathbb{N})$ and X are in bijection

Define $\Phi : \mathcal{P}(\mathbb{N}) \to X$ by

$$\Phi(A) = \{\omega_i\}_{i\in\mathbb{N}}, \qquad \omega_i = \mathbb{1}\{i \in A\}, \quad \forall A \in \mathcal{P}(\mathbb{N}),$$

i.e. $\Phi(A)$ is the indicator sequence of A: coordinate i is 1 if $i \in A$ and 0 otherwise. We show Φ is a bijection.

- **Injective:** if $A \neq B$, then without loss of generality there is some $i \in A \setminus B$, so $\Phi(A)_i = 1 \neq 0 = \Phi(B)_i$, hence $\Phi(A) \neq \Phi(B)$.
- **Surjective:** given any $\omega = (\omega_i)_{i\in\mathbb{N}} \in X$, let $A = \{i \in \mathbb{N} : \omega_i = 1\} \in \mathcal{P}(\mathbb{N})$. Then $\Phi(A) = \omega$ by construction.

Hence Φ is a bijection, so $|\mathcal{P}(\mathbb{N})| = |X|$.

Conclusion

By Part (2), Φ is a bijection, so $|\mathcal{P}(\mathbb{N})| = |X|$. Since bijections preserve cardinality (and X is uncountable by Part (1)), it follows that $\mathcal{P}(\mathbb{N})$ is uncountable as well.//

Q3: A.9

Apply the principle of induction to establish the following:

(a)

For each $n \in \mathbb{N}$, $\sum_{j=1}^{n} j^2 = \dfrac{n(n+1)(2n+1)}{6}$.

Answer

Proof. Let $P(n)$ be the statement

$$\sum_{j=1}^{n} j^2 = \frac{n(n+1)(2n+1)}{6}.$$

Base case ($n = 1$): The left side is $\sum_{j=1}^{1} j^2 = 1$. The right side is $\dfrac{1 \cdot 2 \cdot 3}{6} = 1$. Since equality holds, the base case $P(1)$ holds.

Inductive step: Let $n \in \mathbb{N}$ be given, and suppose $P(n)$ holds, i.e.

$$\sum_{j=1}^{n} j^2 = \frac{n(n+1)(2n+1)}{6}.$$

Then consider $P(n+1)$. By the inductive hypothesis

$$\sum_{j=1}^{n+1} j^2 = \sum_{j=1}^{n} j^2 + (n+1)^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2,$$

Factoring out $(n+1)$,

$$\sum_{j=1}^{n+1} j^2 = (n+1)\left[\frac{n(2n+1)}{6} + (n+1)\right] = (n+1) \cdot \frac{n(2n+1) + 6(n+1)}{6} = (n+1) \cdot \frac{2n^2 + 7n + 6}{6}.$$

Simplifying, using $2n^2 + 7n + 6 = (n+2)(2n+3)$, gives

$$\sum_{j=1}^{n+1} j^2 = \frac{(n+1)(n+2)(2n+3)}{6} = \frac{(n+1)\big((n+1)+1\big)\big(2(n+1)+1\big)}{6},$$

which is exactly the statement $P(n+1)$.

Thus $P(n) \implies P(n+1)$. By induction, $P(n)$ holds for all $n \in \mathbb{N}$.//

(b)

For each $n \in \mathbb{N}$, $x_1, x_2, \ldots, x_k \in \mathbb{R}$,

(i)

(The binomial formula). $(x_1 + x_2)^n = \sum_{r=0}^{n} \binom{n}{r} x_1^r x_2^{n-r}$.

Answer

Proof. Let $P(n)$ be the statement

$$(x_1 + x_2)^n = \sum_{r=0}^{n} \binom{n}{r} x_1^r x_2^{n-r}.$$

Base case ($n = 1$): The left side is $(x_1 + x_2)^1 = x_1 + x_2$. The right side is

$$\sum_{r=0}^{1} \binom{1}{r} x_1^r x_2^{1-r} = \binom{1}{0} x_2 + \binom{1}{1} x_1 = x_1 + x_2.$$

Since equality holds, the base case $P(1)$ holds.

Inductive step: Let $n \in \mathbb{N}$ be given, and suppose $P(n)$ holds. We then consider $P(n+1)$. By the inductive hypothesis

$$(x_1 + x_2)^{n+1} = (x_1 + x_2)(x_1 + x_2)^n = (x_1 + x_2) \sum_{r=0}^{n} \binom{n}{r} x_1^r x_2^{n-r}$$

Distributing,

$$(x_1 + x_2)^{n+1} = \sum_{r=0}^{n} \binom{n}{r} x_1^{r+1} x_2^{n-r} + \sum_{r=0}^{n} \binom{n}{r} x_1^r x_2^{n-r+1}.$$

Re-indexing the first sum with $r \to r - 1$,

$$(x_1 + x_2)^{n+1} = \sum_{r=1}^{n+1} \binom{n}{r-1} x_1^r x_2^{n+1-r} + \sum_{r=0}^{n} \binom{n}{r} x_1^r x_2^{n+1-r}.$$

Separating out the $r = 0$ term from the first sum's absence and the $r = n + 1$ term from the second sum's absence, and combining,

$$(x_1 + x_2)^{n+1} = x_2^{n+1} + \sum_{r=1}^{n} \left[\binom{n}{r-1} + \binom{n}{r} \right] x_1^r x_2^{n+1-r} + x_1^{n+1}.$$

By Pascal's Rule, $\binom{n}{r-1} + \binom{n}{r} = \binom{n+1}{r}$, so

$$(x_1 + x_2)^{n+1} = \binom{n+1}{0} x_2^{n+1} + \sum_{r=1}^{n} \binom{n+1}{r} x_1^r x_2^{n+1-r} + \binom{n+1}{n+1} x_1^{n+1} = \sum_{r=0}^{n+1} \binom{n+1}{r} x_1^r x_2^{n+1-r},$$

which is exactly the statement $P(n + 1)$.

Thus $P(n) \implies P(n + 1)$. By induction, $P(n)$ holds for all $n \in \mathbb{N}$.//

(ii)

(The multinomial formula).

$$(x_1 + x_2 + \ldots + x_k)^n = \sum \frac{n!}{r_1! r_2! \ldots r_k!} x_1^{r_1} x_2^{r_2} \ldots x_k^{r_k},$$

where the summation extends over all $(r_1, r_2, \ldots, r_k)$ such that $r_i \in \mathbb{N}$, $0 \leq r_i \leq n$, $\sum_{i=1}^{k} r_i = n$.

Answer

Proof. Fix $k \in \mathbb{N}$ and $x_1, \ldots, x_k \in \mathbb{R}$. Let $P(n)$ be the statement

$$(x_1 + x_2 + \cdots + x_k)^n = \sum \frac{n!}{r_1! r_2! \cdots r_k!} x_1^{r_1} x_2^{r_2} \cdots x_k^{r_k},$$

where the sum extends over all $(r_1, \ldots, r_k)$ such that $r_i \in \mathbb{N} \cup \{0\}$, $0 \leq r_i \leq n$, and $\sum_{i=1}^{k} r_i = n$.

Base case ($n = 1$): The left side is $x_1 + x_2 + \cdots + x_k$. On the right side, since $\sum_{i=1}^{k} r_i = 1$ and each $r_i \geq 0$, exactly one r_i equals 1 and all others equal 0. The term corresponding to $r_i = 1$ is

$$\frac{1!}{0! \cdots 1! \cdots 0!} x_1^0 \cdots x_i^1 \cdots x_k^0 = x_i,$$

so summing over $i = 1, \dots, k$ gives $x_1 + x_2 + \cdots + x_k$. Since equality holds, the base case $P(1)$ holds.

Inductive step: Let $n \in \mathbb{N}$ be given, and suppose $P(n)$ holds. We then turn to $P(n+1)$. By the inductive hypothesis

$$(x_1 + \cdots + x_k)^{n+1} = (x_1 + \cdots + x_k)^n (x_1 + \cdots + x_k) = \left(\sum_{r_1 + \cdots + r_k = n} \frac{n!}{r_1! \cdots r_k!} x_1^{r_1} \cdots x_k^{r_k} \right) (x_1 + \cdots + x_k)$$

Distributing over the k terms $x_1, \dots, x_k$,

$$(x_1 + \cdots + x_k)^{n+1} = \sum_{i=1}^{k} \sum_{r_1 + \cdots + r_k = n} \frac{n!}{r_1! \cdots r_k!} x_1^{r_1} \cdots x_i^{r_i + 1} \cdots x_k^{r_k}.$$

For fixed i, re-index by setting $s_j = r_j$ for $j \neq i$ and $s_i = r_i + 1$, so that $\sum_{j=1}^{k} s_j = n + 1$ and $s_i \geq 1$. Since $r_i = s_i - 1$ gives $(s_i - 1)! = s_i!/s_i$,

$$\frac{n!}{r_1! \cdots r_i! \cdots r_k!} = \frac{n!}{s_1! \cdots (s_i - 1)! \cdots s_k!} = \frac{n!\, s_i}{s_1! \cdots s_i! \cdots s_k!}.$$

Since the $s_i = 0$ terms vanish anyway, we can drop the restriction $s_i \geq 1$ and sum over the full range $s_1 + \cdots + s_k = n + 1$:

$$(x_1 + \cdots + x_k)^{n+1} = \sum_{i=1}^{k} \sum_{s_1 + \cdots + s_k = n+1} \frac{n!\, s_i}{s_1! \cdots s_k!} x_1^{s_1} \cdots x_k^{s_k}.$$

Swapping the order of summation and collecting terms with the same exponents $(s_1, \dots, s_k)$,

$$(x_1 + \cdots + x_k)^{n+1} = \sum_{s_1 + \cdots + s_k = n+1} \frac{n!}{s_1! \cdots s_k!} \left(\sum_{i=1}^{k} s_i \right) x_1^{s_1} \cdots x_k^{s_k}.$$

Since $\sum_{i=1}^{k} s_i = n + 1$ for every term in the sum, this is

$$(x_1 + \cdots + x_k)^{n+1} = \sum_{s_1 + \cdots + s_k = n+1} \frac{(n+1)!}{s_1! \cdots s_k!} x_1^{s_1} \cdots x_k^{s_k},$$

which is exactly the statement $P(n+1)$.

Thus $P(n) \implies P(n+1)$. By induction, $P(n)$ holds for all $n \in \mathbb{N}$.//

Q4: A.11

Show that the set $A = \{r : r \in \mathbb{Q}, r^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no l.u.b. in $\mathbb{Q}$.

Answer

Sketch: We establish (1): A is bounded above, and (2): No rational M can be the *least* upper bound, by supposing $M = \sup_{\mathbb{Q}}(A)$ exists and deriving a contradiction (two contradictions, one for each case).

(1): A is bounded above

Proof. Let $r \in A$ be given. Thus $r^2 < 2$. If $r \geq 2$, then $r^2 \geq 4 > 2$, contradicting $r \in A$. Thus $r < 2$. By definition of an upper bound, 2 is an upper bound of A.

(2): A has no l.u.b. in $\mathbb{Q}$

Proof. Suppose, for contradiction, that $M = \sup_{\mathbb{Q}}(A)$ exists. Since $1 \in A$ (as $1^2 = 1 < 2$), $M \geq 1$, so M is a positive rational. Since $\sqrt{2}$ is irrational, no rational squares to 2, so $M^2 \neq 2$. Thus either $M^2 < 2$ or $M^2 > 2$; we then examine each case.

Case $M^2 < 2$: Let $\varepsilon = 2 - M^2 > 0$. Choose a rational δ with $0 < \delta < \min\left(1, \dfrac{\varepsilon}{2M+1}\right)$, and set $r = M + \delta$. Then

$$r^2 = M^2 + 2M\delta + \delta^2 < M^2 + \delta(2M+1) < M^2 + \varepsilon = 2.$$

So $r \in A$, yet $r = M + \delta > M$, contradicting that M is an upper bound of A.

Case $M^2 > 2$: Let $\varepsilon = M^2 - 2 > 0$. Choose a rational δ with $0 < \delta < \min\left(M, \dfrac{\varepsilon}{2M}\right)$, and set $b = M - \delta$. Then

$$b^2 = M^2 - 2M\delta + \delta^2 > M^2 - 2M\delta > M^2 - \varepsilon = 2.$$

Since $b > 0$ and $b^2 > 2$: for any $a \in A$, $a^2 < 2 < b^2$, and since $a, b > 0$, this gives $a < b$. So b is an upper bound of A, yet $b < M$, contradicting that M is the *least* upper bound.

Both cases are contradictions, leading to the conclusion that no such $M \in \mathbb{Q}$ exists. Thus A has no l.u.b. in $\mathbb{Q}$.//

Q5: A.12

Show that for any two sequences $\{x_n\}_{n\geq 1}, \{y_n\}_{n\geq 1} \subset \mathbb{R}$,

$$\varliminf_{n\to\infty} x_n + \varliminf_{n\to\infty} y_n \leq \varliminf_{n\to\infty} (x_n + y_n) \leq \varlimsup_{n\to\infty} (x_n + y_n)$$

$$\leq \varlimsup_{n\to\infty} x_n + \varlimsup_{n\to\infty} y_n.$$

Answer

Proof: By definition, for every $n \in \mathbb{N}$ and every $k \geq n$,

$$\inf_{k\geq n} x_k \leq x_k \leq \sup_{k\geq n} x_k, \qquad \inf_{k\geq n} y_k \leq y_k \leq \sup_{k\geq n} y_k.$$

Taking $k = n$ and adding the left- and right-hand inequalities gives, for every n,

$$\inf_{k\geq n} x_k + \inf_{k\geq n} y_k \ \leq\ x_n + y_n \ \leq\ \sup_{k\geq n} x_k + \sup_{k\geq n} y_k. \tag{1}$$

Since $\inf_{k\geq n} x_k$ is non-decreasing in n and $\sup_{k\geq n} x_k$ is non-increasing in n (and likewise for y_n), both are monotone sequences and hence by definition converge to $\varliminf_{n\to\infty} x_n$ and $\varlimsup_{n\to\infty} x_n$ respectively. So the left side of (1) converges, as $n \to \infty$, to $\underline{\lim}_{n\to\infty} x_n + \underline{\lim}_{n\to\infty} y_n$, and the right side converges to $\overline{\lim}_{n\to\infty} x_n + \overline{\lim}_{n\to\infty} y_n$.

Now apply $\liminf$ and $\limsup$ to (1), using that both are monotone with respect to termwise $\leq$. From the left inequality of (1),

$$\underline{\lim}_{n\to\infty} \left(\inf_{k\geq n} x_k + \inf_{k\geq n} y_k \right) \leq \underline{\lim}_{n\to\infty} (x_n + y_n),$$

and since the left side is a limit (shown above), it equals $\underline{\lim}_{n\to\infty} x_n + \underline{\lim}_{n\to\infty} y_n$. This gives

$$\underline{\lim}_{n\to\infty} x_n + \underline{\lim}_{n\to\infty} y_n \ \leq \ \underline{\lim}_{n\to\infty} (x_n + y_n).$$

Symmetrically, applying $\limsup$ to the right inequality of (1) gives

$$\overline{\lim}_{n\to\infty} (x_n + y_n) \ \leq \ \overline{\lim}_{n\to\infty} x_n + \overline{\lim}_{n\to\infty} y_n.$$

Finally, for any real sequence z_n we have $\inf_{k\geq n} z_k \leq \sup_{k\geq n} z_k$ for every n, and taking $n \to \infty$ gives $\underline{\lim}_{n\to\infty} z_n \leq \overline{\lim}_{n\to\infty} z_n$. Applying this with $z_n = x_n + y_n$ gives the middle inequality,

$$\underline{\lim}_{n\to\infty} (x_n + y_n) \leq \overline{\lim}_{n\to\infty} (x_n + y_n).$$

Chaining the three inequalities above gives us:

$$\underline{\lim}_{n\to\infty} x_n + \underline{\lim}_{n\to\infty} y_n \leq \underline{\lim}_{n\to\infty} (x_n + y_n) \leq \overline{\lim}_{n\to\infty} (x_n + y_n) \leq \overline{\lim}_{n\to\infty} x_n + \overline{\lim}_{n\to\infty} y_n.//$$

Q6: A.17

Find the radius of convergence, ρ, for the powers series $A(x) \equiv \sum_{n=0}^{\infty} a_n x^n$ where

(a)

$a_n = \dfrac{n}{(n+1)}$, $n \geq 0$.

Answer

Sketch: Since $\rho = \left(\overline{\lim}_{n\to\infty} |a_n|^{1/n}\right)^{-1}$, it suffices to show $a_n^{1/n} \to 1$.

Proof. For $n \geq 1$,

$$\log\left(a_n^{1/n}\right) = \frac{1}{n}\log\left(\frac{n}{n+1}\right) = -\frac{1}{n}\log\left(1 + \frac{1}{n}\right).$$

Writing $t = 1/n \to 0$ and using $\lim_{t\to 0} \dfrac{\log(1+t)}{t} = 1$,

$$\frac{1}{n}\log\left(1 + \frac{1}{n}\right) = \frac{1}{n^2} \cdot \frac{\log(1+1/n)}{1/n} \ \longrightarrow \ 0 \cdot 1 = 0.$$

Hence $\log(a_n^{1/n}) \to 0$, so $a_n^{1/n} \to e^0 = 1$. Since this is an honest limit, $\overline{\lim}_{n\to\infty} |a_n|^{1/n} = 1$.

Thus $\rho = 1^{-1} = 1.//$

(b)

$a_n = n^p$, $n \geq 0$, $p \in \mathbb{R}$.

Answer

Sketch: The key fact is $\log n/n \to 0$, which gives $n^{p/n} \to 1$ for every fixed p, so $\rho = 1$ regardless of the exponent.

Proof. For the continuous function $f(x) = \dfrac{\log x}{x}$, L'Hopital's rule gives

$$\lim_{x\to\infty} \frac{\log x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = 0,$$

and since this limit holds as $x \to \infty$ through the reals, it holds along the integer sequence: $\dfrac{\log n}{n} \to 0$.

For $n \geq 1$,

$$\log\left(a_n^{1/n}\right) = \log\left(n^{p/n}\right) = p \cdot \frac{\log n}{n} \longrightarrow p \cdot 0 = 0$$

for every fixed $p \in \mathbb{R}$. Hence $a_n^{1/n} = n^{p/n} \to e^0 = 1$, so $\overline{\lim}_{n\to\infty} |a_n|^{1/n} = 1$.

(For $p < 0$ the term $a_0 = 0^p$ is undefined, but the radius of convergence depends only on the tail behavior as $n \to \infty$, so this does not affect ρ.)

Thus $\rho = 1^{-1} = 1$, for every $p \in \mathbb{R}$.//

(c)

$a_n = \dfrac{1}{n!}$, $n \geq 0$ (where $0! = 1$).

Answer

Sketch: We show $(1/n!)^{1/n} \to 0$ using the elementary bound $n! \geq (n/e)^n$, obtained by truncating the series for e^n to a single term (without invoking Stirling's formula, though that is also an option).

Proof. For $n \geq 1$, the exponential series $e^n = \displaystyle\sum_{k=0}^{\infty} \frac{n^k}{k!}$ has all nonnegative terms, so it is bounded below by its $k = n$ term:

$$e^n \geq \frac{n^n}{n!} \quad \Longrightarrow \quad n! \geq \frac{n^n}{e^n} = \left(\frac{n}{e}\right)^n.$$

Taking n-th roots (all quantities positive),

$$(n!)^{1/n} \geq \frac{n}{e}.$$

Hence

$$0 < \left(\frac{1}{n!}\right)^{1/n} = \frac{1}{(n!)^{1/n}} \leq \frac{e}{n} \longrightarrow 0.$$

By the squeeze (or sandwich, they are one-and-the-same) theorem, $\lim_{n\to\infty}\left(\frac{1}{n!}\right)^{1/n} = 0$, so $\overline{\lim}_{n\to\infty} |a_n|^{1/n} = 0$.

Thus $\rho = (0)^{-1} = +\infty$: the series converges for every $x \in \mathbb{R}$.//

Q7: A.18

(a)

Find the Taylor series at $a = 0$ for the function $f(x) = \frac{1}{1-x}$ in $I \equiv (-1, +1)$ and show that it converges to $f(x)$ on I.

Answer

Sketch: We utilize a proof by induction to find a closed form for $f^{(n)}(x)$. We then evaluate the series.

Claim. For all $n \in \mathbb{N} \cup \{0\}$, $f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$.

Proof. Base case ($n = 0$): $f^{(0)}(x) = f(x) = \frac{1}{1-x} = \frac{0!}{(1-x)^{0+1}}$, so the claim holds.

Inductive step: Suppose $f^{(n)}(x) = \frac{n!}{(1-x)^{n+1}}$ for some $n \geq 0$.

Differentiating with respect to x,

$$f^{(n+1)}(x) = \frac{d}{dx}\left[n!(1-x)^{-(n+1)}\right] = n!\left(-(n+1)(1-x)^{-(n+2)}\right)\cdot(-1) = \frac{(n+1)!}{(1-x)^{(n+1)+1}},$$

establishing the claim holds for $n+1$. So by induction, the formula holds for all $n \in \mathbb{N} \cup \{0\}$.

The Taylor series at $a = 0$. Since $f^{(n)}(0) = n!$ for all n, the Taylor series is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}(x-0)^n = \sum_{n=0}^{\infty} \frac{n!}{n!}x^n = \sum_{n=0}^{\infty} x^n.$$

Convergence to f on $I = (-1, 1)$. For $x \in I$, let $S_N(x) = \sum_{n=0}^{N} x^n$ be the N-th partial sum. By the finite geometric sum formula (since $x \neq 1$),

$$S_N(x) = \frac{1 - x^{N+1}}{1-x} = \frac{1}{1-x} - \frac{x^{N+1}}{1-x}.$$

Since $|x| < 1$, we have $x^{N+1} \to 0$ as $N \to \infty$, so

$$\lim_{N\to\infty} S_N(x) = \frac{1}{1-x} - 0 = f(x).$$

Hence $\sum_{n=0}^{\infty} x^n$ converges to $f(x)$ for every $x \in I = (-1, 1)$.//

(b)

Find the Taylor series of $1 + x + x^2$ in $I = (1, 3)$, centered at 2.

Answer

Sketch: Since g is a degree-2 polynomial, we need only compute g, g', g'' at $a = 2$ and assemble the finite sum, and then verify that it reproduces $g(x)$.

Let $g(x) = 1 + x + x^2$, and consider its Taylor expansion centered at $a = 2$.

Derivatives. We compute

$$g(x) = 1 + x + x^2, \qquad g'(x) = 1 + 2x, \qquad g''(x) = 2, \qquad g^{(n)}(x) = 0 \text{ for } n \geq 3.$$

Evaluating at $a = 2$:

$$g(2) = 1 + 2 + 4 = 7, \qquad g'(2) = 1 + 4 = 5, \qquad g''(2) = 2, \qquad g^{(n)}(2) = 0 \ (n \geq 3).$$

Taylor series. Since all derivatives of order ≥ 3 vanish, the Taylor series at $a = 2$ is the finite sum

$$g(x) = \sum_{n=0}^{\infty} \frac{g^{(n)}(2)}{n!}(x-2)^n = g(2) + g'(2)(x-2) + \frac{g''(2)}{2!}(x-2)^2 = 7 + 5(x-2) + (x-2)^2.$$

Verification. Expanding,

$$7 + 5(x-2) + (x-2)^2 = 7 + 5x - 10 + x^2 - 4x + 4 = x^2 + x + 1,$$

which recovers $g(x) = 1 + x + x^2$ exactly. Since this is a finite sum, it holds for all $x \in \mathbb{R}$, and in particular on the interval $I = (1, 3)$.//

(c)

Let

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } |x| < 1, x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

(i)

Show that f is infinitely differentiable at 0 and compute $f^{(j)}(0)$ for all $j \geq 1$.

Answer

Sketch: There are two parts to this proof: (1) that f is infinitely differentiable, using the chain rule for derivation, and (2) compute $f^{(j)}(0)$ utilizing an inductive proof.

Setup. For $x \neq 0$ (with $|x| < 1$) and $j \geq 1$, each derivative of f has the form

$$f^{(j)}(x) = \sum_{k=0}^{j-1} a_{k,j}\, x^{2k-3j}\, e^{-1/x^2}, \qquad x \neq 0,$$

for suitable constants $a_{k,j}$ (this follows by repeated differentiation via the chain and product rules; each differentiation multiplies existing terms by a rational function of $1/x$ and adds new terms of the same general shape). We take this representation as given and focus on differentiability at $x = 0$.

Lemma. For every integer $m \geq 0$,

$$\lim_{x\to 0} \frac{e^{-1/x^2}}{x^m} = 0.$$

Proof. Substituting $s = 1/|x|$ so that $s \to \infty$ as $x \to 0$, we have $\left|\frac{e^{-1/x^2}}{x^m}\right| = s^m e^{-s^2} \to 0$ as $s \to \infty$, since the exponential e^{-s^2} decays faster than any polynomial power s^m grows.

Proof (by induction on j) that $f^{(j)}(0)$ exists and equals 0 for all $j \geq 1$.

Note first that f is continuous at 0: $\lim_{x\to 0} f(x) = \lim_{x\to 0} e^{-1/x^2} = 0$ by the Lemma given above (with $m = 0$), which agrees with $f(0) = 0$.

Base case ($j = 1$): By the definition of the derivative (at least as defined in the Appendix of the text),

$$f'(0) = \lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{e^{-1/h^2}}{h},$$

which equals 0 by the Lemma (with $m = 1$). So $f'(0)$ exists and equals 0.

Inductive step: Suppose $j \geq 1$, $f^{(j)}(0) = 0$, and, for $x \neq 0$, $f^{(j)}(x) = \sum_{k=0}^{j-1} a_{k,j} x^{2k-3j} e^{-1/x^2}$ (as given in the Setup, valid since $j \geq 1$). Then by the definition of the derivative,

$$f^{(j+1)}(0) = \lim_{h\to 0} \frac{f^{(j)}(h) - f^{(j)}(0)}{h} = \lim_{h\to 0} \frac{f^{(j)}(h)}{h} = \lim_{h\to 0} \sum_{k=0}^{j-1} a_{k,j} \frac{e^{-1/h^2}}{h^{3j-2k+1}}.$$

Each term in this finite sum tends to 0 by the Lemma (with $m = 3j - 2k + 1 \geq 0$). Hence the whole limit is 0, so $f^{(j+1)}(0)$ exists and equals 0.

By induction, $f^{(j)}(0)$ exists and equals 0 for every $j \geq 1$; together with $f(0) = 0$, f is infinitely differentiable at 0 with $f^{(j)}(0) = 0$ for all $j \geq 1$.//

(ii)

Show that the Taylor series at $a = 0$ converges but not to f on $(-1, 1)$.

Answer

Sketch: We can apply the result from part (c)(i). Then, since $f(x) = e^{-1/x^2} > 0$ for every $x \neq 0$, the series (identically 0) converges to a value different from $f(x)$ at every nonzero point of $(-1, 1)$.

Proof. From part (c)(i), $f^{(n)}(0) = 0$ for all $n \geq 0$ (including $f(0) = 0$). Therefore the Taylor series at $a = 0$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}(x-0)^n = \sum_{n=0}^{\infty} \frac{0}{n!} x^n = \sum_{n=0}^{\infty} 0 = 0,$$

i.e., the Taylor series is the zero function. This series converges (trivially to 0) for every $x \in \mathbb{R}$, and in particular for every $x \in (-1, 1)$.

However, for any $x \in (-1, 1)$ with $x \neq 0$, we have $f(x) = e^{-1/x^2} > 0 \neq 0$. Hence the Taylor series converges to 0, but $f(x) \neq 0$, so the Taylor series does **not** converge to $f(x)$ at any such point. And only at $x = 0$ does the series' value (0) agree with $f(0) = 0$.

Thus the Taylor series of f at $a = 0$ converges on all of $(-1, 1)$, but it converges to f only at the single point $x = 0$, not on $(-1, 1)$ as a whole.//

Q8: A.24 (c)

Draw the open unit ball $B_p \equiv \{x : x \in \mathbb{R}^2,\ d_p(x, 0) < 1\}$ in $\mathbb{R}^2$ for $p = 1, 2$ and ∞.

Answer

$p = 1, 2$ and ∞ are shown below in black, red, and blue respectively.